GGG-B-490E
24 September 1984
SUPERSEDING
GGG-B-490D
June 7, 1968

FEDERAL SPECIFICATION

BLOCKS, TACKLE (MANILA AND NYLON ROPE)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single, double, and multiple sheave tackle blocks for use with manila and nylon rope.

1.2 Classification. The blocks shall be of the following types, and shall have the number of sheaves, as specified (see 6.2):

Type I - Nontoppling blocks.
  Double sheave.
  Triple sheave.
Type II- Ordinary blocks.
  Single sheave.
  Double sheave.
  Triple sheave.
Type IIa - Blocks, tackle (awning pulley).
  Single sheave.
  Double sheave.
Type III - Snatch blocks, safety locking.
  Single sheave.
Type IV - Safety blocks.
  Double sheave.

FSC 3940

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-C-271 - Chains and Attachments, Welded and Weldless.
TT-V-109 - Varnish, Interior, Alkyd-Resin.

Federal Standards:

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-123	- Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal Specifications, and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-B-3865	- Blocks, Rope, Tackle: Packaging of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

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Military Standards:

MIL-STD-22	- Welded-Joint Design.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-889	- Dissimilar Metals.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations:

41 CFR 1-1.25	- Recovered Materials.

(The Code of Federal Regulations (CFR) is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or solicitation for offers, shall apply.

American Society for Testing and Materials (ASTM):

A 153 - Zinc Coating (Hot Dip) on Iron and Steel Hardware.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

American Iron and Steel Institute (AISI):

Steel Products Manual - Carbon Steel: Semi-Finished for Forgings.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

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3. REQUIREMENTS

3.1 Description. The block shall consist of one or more sheaves, a center pin, shells with or without straps, and a rig assembly as shown in figures 1 through 6 (see 6.4).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be new and unused.

3.3.1 Material deterioration and control. The blocks shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practicable.

3.4 Strength. The safe working load (SWL) shall be the maximum load that may be imposed on the complete block, including the rig assemblies (except the hooks), while maintaining a safety factor of 5. Hooks shall have a minimum safety factor of 4. Blocks shall be stable and shall distribute the load to the various components.

3.4.1 Proof load. When subjected to a proof load of twice the SWL specified in the applicable table, the block and fittings shall show no evidence of deformation, distortion, cracks, or permanent set.

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3.4.2 Ultimate strength. When tested for ultimate strength, the hook shall fail at not less than 4 times the applicable SWL of the block. The block and fittings shall fail at not less than 5 times the applicable SWL. When tested as specified in 4.6.2, the hook shall fail before failure of the block or the fittings.

3.5 Threaded parts. Threaded parts shall conform to FED-STD-H28.

3.6 Sheaves. Sheaves shall be gray iron, malleable or ductile cast iron, or cast steel and shall be designed to prevent cutting action on the rope. Sheaves shall be designed so that only the hubs bear against the shells or straps. Sheave rims shall be grooved to fit the specified rope size. Overall sideplay shall be not more than 1/16 inch. Dimensions of the sheaves shall be as specified in the applicable table.

3.6.1 Bushings. Each sheave shall be furnished with a one-piece, oil-self-lubricating, high-grade bronze bushing which shall be pressed into the sheave with sufficient tightness to prevent slippage at loads of four times the SWL. The bushing shall be equal in length to the sheave thickness at the hub and shall have a wall thickness not less than 3/16 inch.

3.7 Center pin. The center pin shall not rotate and shall transmit the sheave load to the straps. The pin shall be AISI 1040 steel (80,000 pounds per square inch (psi) minimum ultimate tensile strength and 45,000 psi yield, normalized after machining).

3.8 Fastening devices. All pins, bolts, and similar parts shall be adjustable. Such parts shall not be swaged, peened, staked, or otherwise permanently deformed.

3.9 Rig assemblies. Blocks shall be fitted with rig assemblies conforming to the rig assemblies shown in figure 1 and shall be as specified herein. All rig assemblies shall be of forged steel.

3.10 Finish. All metal surfaces of the block assemblies, except bearing surfaces and bushings, shall be hot-dipped galvanized in accordance with ASTM A 153. Running surfaces and guide surfaces shall be free of irregular coating distribution.

3.11 Identification marking. The block assembly shall be permanently and legibly marked with the manufacturer's name or trademark. In addition, the SWL shall be permanently marked or stamped on the block. For military agencies, the block assembly shall be identified in accordance with MIL-STD-130.

3.12 Type I blocks. Type I blocks shall be as shown in figure 2 and shall conform to the requirements specified in table I. The shell length shall be as specified (see 6.2).

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TABLE I. Type I blocks.

Shell length, minimum	SWL		Rope size	Sheave size		
	Double sheave	Triple sheave	Diameter (approx)	Outside diameter, minimum	Thickness at rim, minimum	Center pin diameter, minimum
Inches	Pounds	Pounds	Inches	Inches	Inches	Inch
4	1,600	2,000	1/2	2-1/4	5/8	3/8
6	3,300	4,000	3/4	3-1/2	1	1/2
8	5,100	6,800	1	4-1/2	1-3/8	5/8
10	7,700	10,000	1-1/4	6-1/4	1-1/2	3/4
12	10,600	14,000	1-1/2	8	1-5/8	3/4
14	15,000	20,000	1-3/4	9-1/2	1-7/8	7/8

3.12.1 Shells. Shells shall be select grade, kiln or air dried birch, beech, or hard maple. Cheeks, centerpieces, and spreaders comprising the shell shall be fastened together to form a strong, rigid unit. Shells of heavy, wide, mortised wood blocks shall be cross bolted. The shells shall carry no part of the load, but shall be designed to protect the sheaves.

3.12.2 Finish. Completed shells shall be thoroughly cleaned and sanded; dipped in high-grade clear lacquer or varnish, provided it is at least equivalent in water resistance to TT-V-109; and when dry, redipped to insure thorough sealing and protection of exposed surfaces against moisture and weather.

3.12.3 Straps. Straps shall support the center pin on both sides of each sheave and shall transmit sheave loads to the fitting (hook, shackle, or other like fitting). Inner and outer straps shall be made of one piece each.

3.12.4 Center pin. The center pin shall be as specified in 3.7, except one end shall have an eye with a thimble through the eye.

3.12.5 Rig assemblies. The block shall be equipped with a swivel hook (rig No. 14) or a swivel releasing hook (rig No. 15), see figure 1, as specified (see 6.2). The body of the hook swivel shall be provided with a recessed hydraulic grease fitting 1/8 inch N.P.T. conforming to MS15003, (see figure 2).

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3.12.6 Rollers. Guide rollers shall be provided in the upper four corners as shown in figure 2.

3.13 Type II blocks. Type II blocks shall be as shown in figure 3 and shall conform to the requirements specified in table II. The shell length shall be as specified (see 6.2).

TABLE II. Type II blocks.

Shell length minimum	SWL			Rope size	Sheave size		
	Single sheave	Double sheave	Triple sheave	Diameter (approx)	Outside diameter, minimum	Thickness at rim, minimum	Center pin diameter, minimum
Inches	Pounds	Pounds	Pounds	Inches	Inches	Inches	Inch
4	1,000	1,600	2,000	1/2	2-1/4	5/8	3/8
6	2,000	3,300	4,000	3/4	3-1/2	1	1/2
8	3,300	5,100	6,800	1	4-1/2	1-3/8	5/8
10	5,000	7,700	10,000	1-1/4	6-1/4	1-1/2	3/4
12	6,800	10,600	14,000	1-1/2	8	1-5/8	3/4
14	9,600	15,000	20,000	1-3/4	9-1/2	1-7/8	7/8

3.13.1. Shells. Shells shall be formed from steel or malleable iron and shall be flanged to provide additional stiffness and reduce wear and chafing of the rope. Shell pieces shall be securely held in position and shall be designed to prevent distortion or spreading under extreme lateral rope pulls.

3.13.2 Straps. Straps shall be as specified in 3.12.3. Straps and shells shall be assembled by means of rivets or by bolts secured with locknuts.

3.13.3 Rig assemblies. Each block shall be fitted with a regular shackle (rig No. 1), and upset shackle (rig No. 2), and upset shackle in loose swivel link (rig No. 4), a loose side hook (rig No. 10), loose sister hooks (rig No. 18), or a swivel hook in loose swivel link (rig No. 11), as shown in figure 1 and as specified (see 6.2).

3.13.3.1 Regular and upset shackles. Regular and upset shackles (figure 1, rigs No. 1 and 2) shall be of the screw-pin type conforming to RR-C-271, type IV, class 1.

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3.13.3.2 Upset shackle in loose swivel link. The shackle shall be fitted to the block by means of a swivel as shown in figure 1, rig No. 4. The shackle shall be free to rotate through 360 degrees, independent of the position of the block. The yield point of the swivel shackle shall be greater than twice the SWL of the block.

3.13.3.3 Loose side hook. The loose side hook shall be as shown in figure 1, rig No. 10, and shall be designed so that the hook will fail by gradual straightening as the ultimate strength of the block is reached.

3.13.3.4 Loose sister hooks. Loose sister hooks shall be fitted to the block by means of loose side eyes as shown in figure 1, rig No. 18.

3.13.3.5 Swivel hook in loose swivel link. The swivel hook shall be fitted to the block by means of a loose swivel link as shown in figure 1, rig No. 11. The hook shall be free to rotate through 360 degrees independent of the position of the block.

3.13.4 Becket. Type II blocks shall be fitted with a becket formed by the extension of two straps below the under side of the block. The becket and the becket pin shall carry the loads specified in table II with a safety factor of 5. A steel thimble of the size specified (see 6.2) shall be fitted over the becket pin.

3.13.5 Type IIa blocks. Each tackle block shall consist of one or two sheaves as specified, sheave pin, and a 360-degree rotation swivel eye. The blocks shall be as shown in figure 4, rig. No. 20, and shall conform to the requirements specified in table III. Shell length shall be as specified (see 6.2).

TABLE III. Type IIa blocks.

Shell length	SWL		Rope Size	Sheave Size	
	Single sheave	Double sheave	Diameter (Nominal)	Diameter (Nominal)	Width (Nominal)
(inches)	(pounds)	(pounds)	(inches)	(inches)	(inches)
3/4	100	150	3/16	3/4	1/4
1	150	200	1/4	1	3/8
1-3/4	250	300	3/8	1-3/4	1/2

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3.13.5.1 Swivel eye. The swivel eye shall be malleable iron or forged steel, and shall be free to swivel. Swivel eye opening shall be the size to accept a shackle or hook of comparable size and strength.

3.13.5.2 Shells. Shells shall be as specified in 3.13.1.

3.14 Type III blocks. Type III blocks shall be as shown in figure 5 and as specified herein. A locking lug on one end of the crosshead shall serve to lock the shackle link in position. The opposite end of the crosshead shall be permanently attached to the swivel link in a manner requiring that the crosshead be rotated 90 degrees from the locked position before releasing the shackle link and opening the block. The connections and fittings shall swing out of position to allow quick entry of rope into the sheave groove without reeving. The crosshead and links shall be forged steel. The blocks shall conform to table IV. The shell length shall be as specified (see 6.2).

TABLE IV. Type III blocks.

Shell length, minimum	SWL	Rope size		Sheave size	
	Single sheave	Diameter (approx)	Outside diameter, minimum	Thickness at rim, minimum	Center pin diameter, minimum
Inches	Pounds	Inches	Inches	Inches	Inch
6	3,000	3/4	3	1-1/8	1/2
8	4,800	1	4-1/2	1-3/8	5/8
10	8,000	1-1/4	5-3/4	1-7/8	3/4
12	10,000	1-1/2	6-3/4	2-1/8	3/4
14	12,000	1-3/4	8	2-1/4	7/8
16	16,000	2	9	2-5/8	1

3.14.1 Shells. Shells shall be as specified in 3.13.1.

3.14.2 Straps. Each strap shall be heavy one-piece construction and shall extend the full length of the shell pieces. Straps shall furnish correct bearing support for the center pin and connections to transmit loads from the sheave to the fitting.

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3.14.3 Rig assemblies. Each block shall be equipped with a swivel hook (rig No. 23), a stiff swivel eye (rig No. 25), or a stiff upset swivel shackle (rig No. 26) as shown in figure 1 and as specified (see 6.2). The rig assemblies shall be attached to the block by a crosshead in which it is free to rotate through 360 degrees. When the block is closed, the rig assembly shall be positioned so that the block is symmetrically loaded.

3.15 Type IV (safety) blocks. Type IV blocks shall be as shown on figure 6 and as specified herein. Unless otherwise specified (see 6.2), blocks shall be provided with a locking mechanism that will grip the rope immediately upon removal of the hoisting strain. The locking mechanism shall cause minimum rope wear and shall securely lock the rope under maximum SWL applicable to the size of the block. The blocks shall be as specified by block number (see 6.2), and shall conform to the requirements specified in table V.

TABLE V. Type IV blocks.

Block size	SWL	Rope size	Sheave size		
	Double sheave	Diameter (approx)	Outside diameter, minimum	Thickness at rim, minimum	Center pin diameter, minimum
	Pounds	Inches	Inches	Inches	Inch
1	600	1/4	1-1/2	3/4	1/4
2	1,250	1/2	2-1/4	5/8	3/8
3	2,500	3/4	3-1/2	1	1/2
4	5,000	1	4-1/2	1-3/8	5/8
5	7,700	1-1/4	6-1/4	1-1/2	3/4

3.15.1 Shells. Shells shall be as specified in 3.13.1, except that for strapless-style blocks the shells shall provide the bearing support for the loads specified in table V. If straps are furnished, the straps shall carry the load and the shells shall carry no part of the load.

3.15.1.1 Shell spacers. Shells shall be spaced at their lower extremities by means of tubing spacers and held in position by rivets or bolts.

3.15.2 Straps. Straps, if used, shall be as specified in 3.13.2.

3.15.3 Center pin. The center pin shall be as specified in 3.7, except that one end shall be hexagon headed and the other end shall have a hexagon locknut or a castellated nut with cotter pin to prevent the nut from loosening.

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3.15.4 Becket. Type IV blocks shall be equipped with a becket as specified in 3.13.4, except that for strapless blocks the becket shall be formed by the extension of the shells.

3.15.5 Rig assemblies. Each block shall be equipped with a loose side hook, figure 1, rig No. 10.

3.16 Workmanship. The blocks shall be free from defects which could affect serviceability.

3.16.1 Castings. Castings shall be uniform quality and free from blow holes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects.

3.16.2 Forgings. Forgings shall be of uniform quality and free from flash, scale, cracks, hard spots, cold shuts, and excessive cooling stresses.

3.16.3 Welding. Welded joint design shall conform to MIL-STD-22. Weld areas shall be free from rust, scale, paint, grease, porosity, splatter, slag, overlap, undercut, nonfusion, cracks, and foreign matter. Welding shall not be used to repair the blocks.

3.16.4 Fastening devices. Screws, pins, bolts, and rivets shall be tight. Parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

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a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Examination. The block shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The block shall be tested as specified in 4.6. Failure of either test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. The acceptable quality (AQL) shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Each block shall be tested as specified in 4.6.1.

4.5 Classification of defects.

4.5.1 Examination. The blocks shall be examined as specified herein for the following defects:

Major

101. Block does not conform to the type or number of sheaves specified.
102. Material not as specified.
103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Components missing; not secured; not positioned, fitted, or assembled as specified.

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107. Rigs and fittings not as specified; inoperative, evidence of binding, misalinement of components or excessive looseness.
108. Threaded parts not as specified.
109. Straps not as specified.
110. Sheaves not as specified.
111. Sheave side play exceeds specified tolerance.
112. Bushing not as specified; loose in sheave, length not as specified, wall thickness not as specified.
113. Center pin not as specified.
114. Locking lug for type III blocks not as specified.
115. Roller guides of type I blocks not positioned as specified.
116. Hook, eye, or shackle does not rotate as specified.
117. Swivel crosshead and link not as specified.
118. Welding not as specified.
119. Fastening devices not as specified.
120. Workmanship not as specified.

Minor

201. Finish not as specified.
202. Identification marking not as specified.

4.6 Tests.

4.6.1 Proof load. Blocks shall be tested in a testing machine equipped with dial gage which will indicate the test load. Pass wire rope of round steel bar loops around the sheave or sheaves and attach to the fixed head of the testing machine. Attach block fitting to the moving head of the machine. Steadily apply load to the block until the applicable proof load specified in 3.4.1 is reached. Nonconformance to 3.4.1 shall constitute failure of this test.

4.6.2 Ultimate load. The first article block, with the applicable hook and fittings shall be installed in a testing machine. A steadily increasing load shall be applied until failure of the hook occurs. The hook shall be removed and the block and fittings shall be further tested to destruction. Any of the following shall constitute failure of this test:

a. The hook fails at less than 4 times the SWL of the block.

b. The block or the fittings fail at less than 5 times the applicable SWL.

c. The block or the fittings fail before the hook fails.

4.7 Inspection of preparation for delivery. The preservation, packing and marking shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-B-3865.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. Blocks shall be preserved and packed in accordance with MIL-B-3865. The degree of preservation shall be level A or commercial and the degree of packing shall be level A, level B, or commercial as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. Packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military activities. Marking for level A or level B shall be in accordance with MIL-STD-129. Marking for commercial shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container for commercial packaging.

6. NOTES

6.1 Intended use. Blocks are intended for use in hoisting and hauling operations. They are general-purpose blocks recognized as a standard product by the tackle industry.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Type of block and number of sheaves required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. Shell length or block number required (see 3.12, 3.13, 3.13.5, 3.14, and 3.15).

e. Rig assembly required for type I blocks (see 3.12.5).

f. Rig assembly required for type II blocks (see 3.13.3).

g. Size of thimble required (see 3.13.4).

h. Rig assembly required for type III blocks (see 3.14.3).

i. When a locking mechanism is not required on type IV blocks (see 3.15).

j. Degree of preservation and degree of packing required (see 5.1).

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6.3 First article. When a first article inspection is required, the item shall be tested and should be a preproduction block. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Figures. The figures show types of blocks which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other blocks which conform to this specification.

MILITARY INTERESTS:

Custodians
Army - ME
Air Force - 99

Review activities
DLA - IS
Air Force - 84

User activities
Navy - SH, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

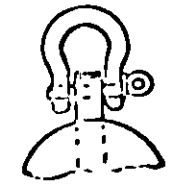
GSA - FSS

PREPARING ACTIVITY:

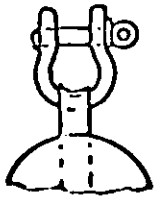
Army - ME

Project 3940-0174

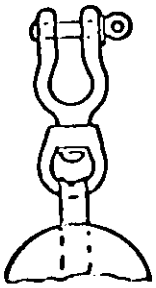
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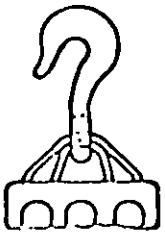
RIG. NO. 1
REGULAR SHACKLE,
LOOSE.



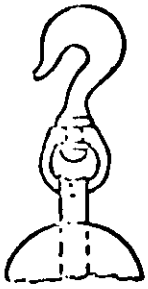
RIG. NO. 2
UPSET SHACKLE,
LOOSE



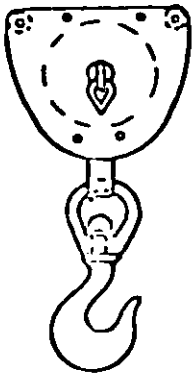
RIG. NO. 4
UPSET SHACKLE IN
LOOSE SWIVEL LINK



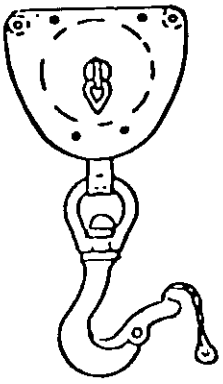
RIG. NO. 10
SIDE HOOK,
LOOSE



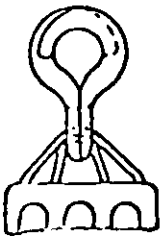
RIG. NO. 11
SINGLE SWIVEL HOOK IN
LOOSE SWIVEL LINK



RIG. NO. 14
SWIVEL HOOK

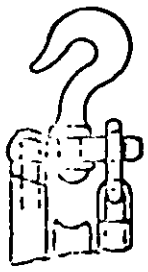


RIG. NO. 15
SWIVEL RELEASING HOOK

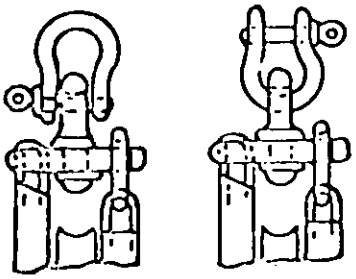


RIG. NO. 18
SIDE SISTER HOOK
LOOSE

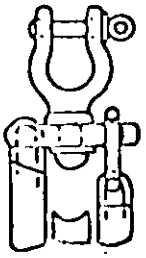
NON-TOPPLING BLOCK WITH LOOSE HOOK.



RIG. NO. 23
STIFF, SWIVEL
HOOK



RIG. NO. 25
REGULAR OR UPSET SHACKLE
IN STIFF SWIVEL EYE.



RIG. NO. 26
STIFF, UPSET
SWIVEL SHACKLE

FIGURE 1. Rigs and fittings for wood and metal rope blocks. (Rigs are identified by numbers only. Each rig consists of a block with fittings assembled as shown.)

X-4113

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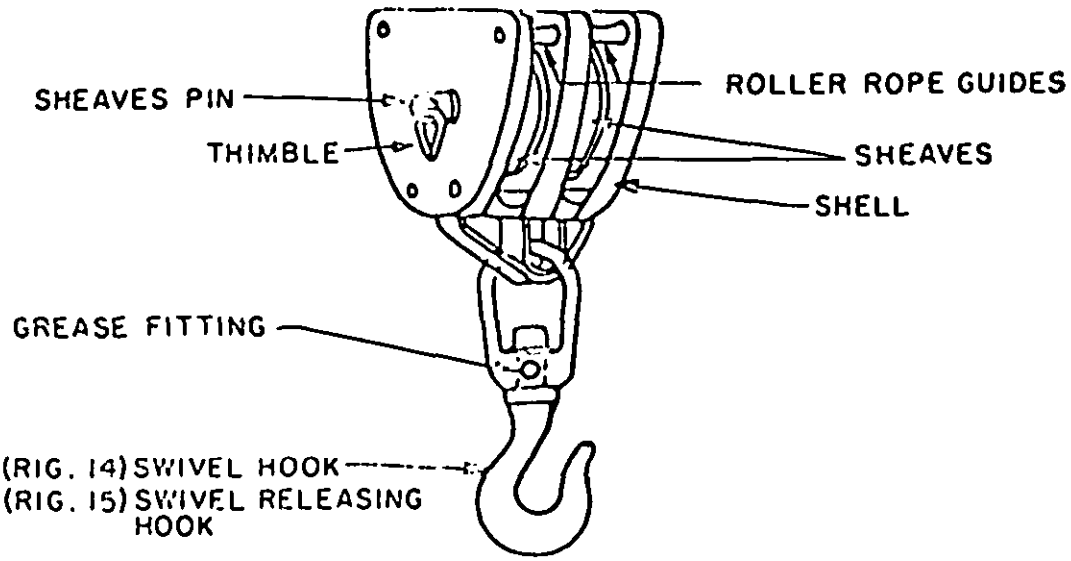


FIGURE 2. Type I, nontoppling block, wood (rig 14 and 15).

X-4114

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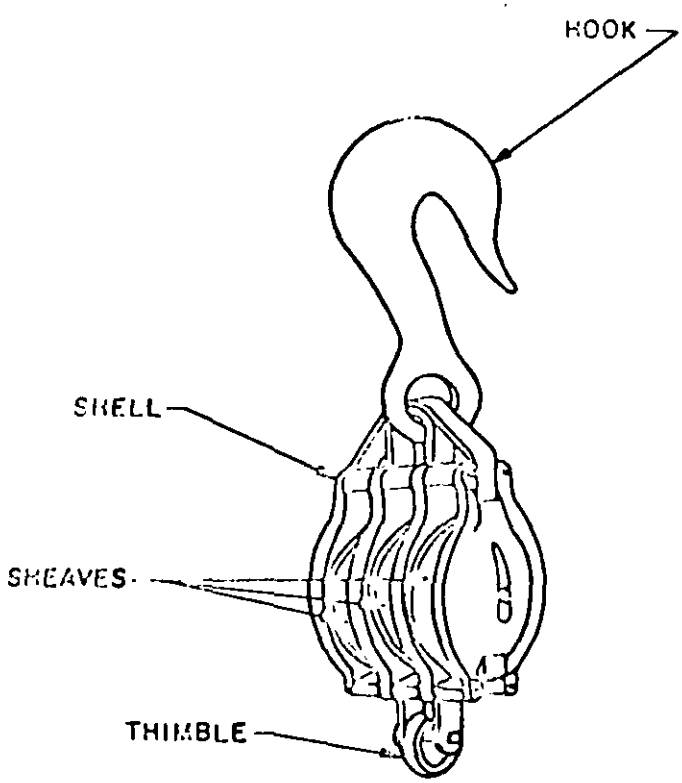


FIGURE 3. Type II, block, steel or malleable iron, ordinary (rig. 10, with bracket shown).

X-4115

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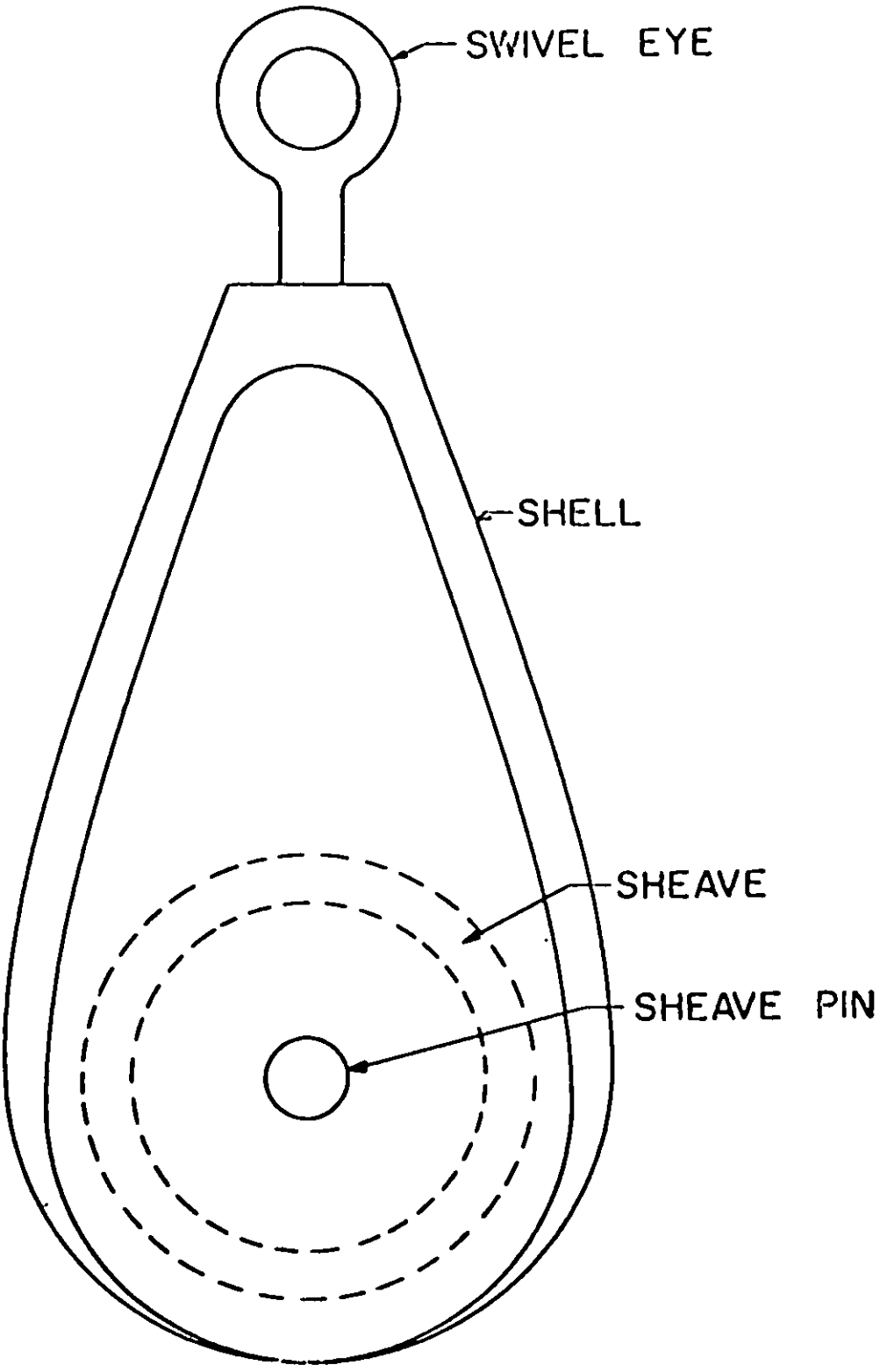


FIGURE 4. Type II a, block, tackle (awning pulley).
(rig. no. 20)

X-2105B

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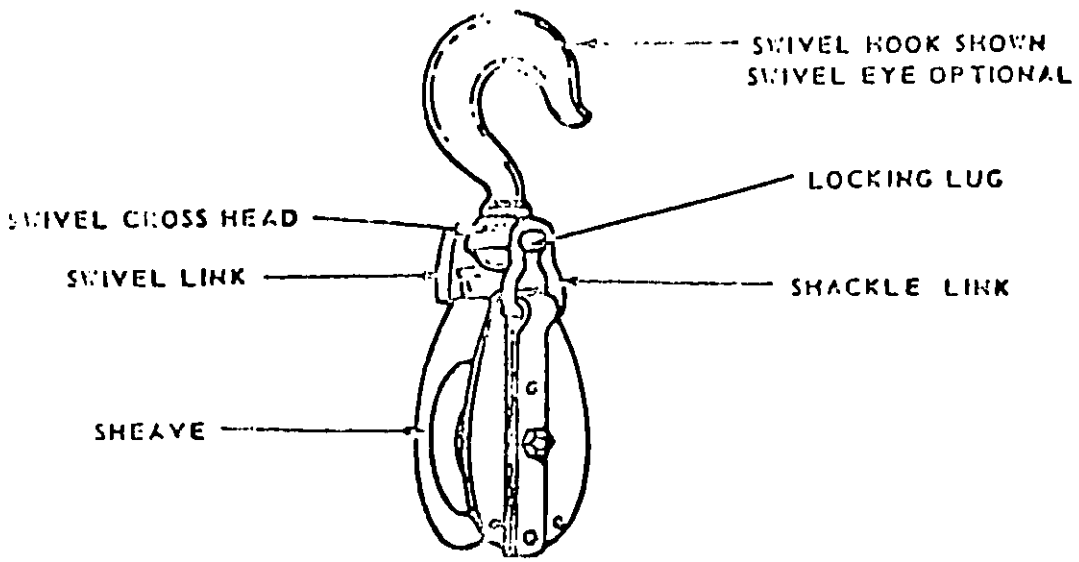


FIGURE 5. Type III, snatch block, safety - locking.

X-4116

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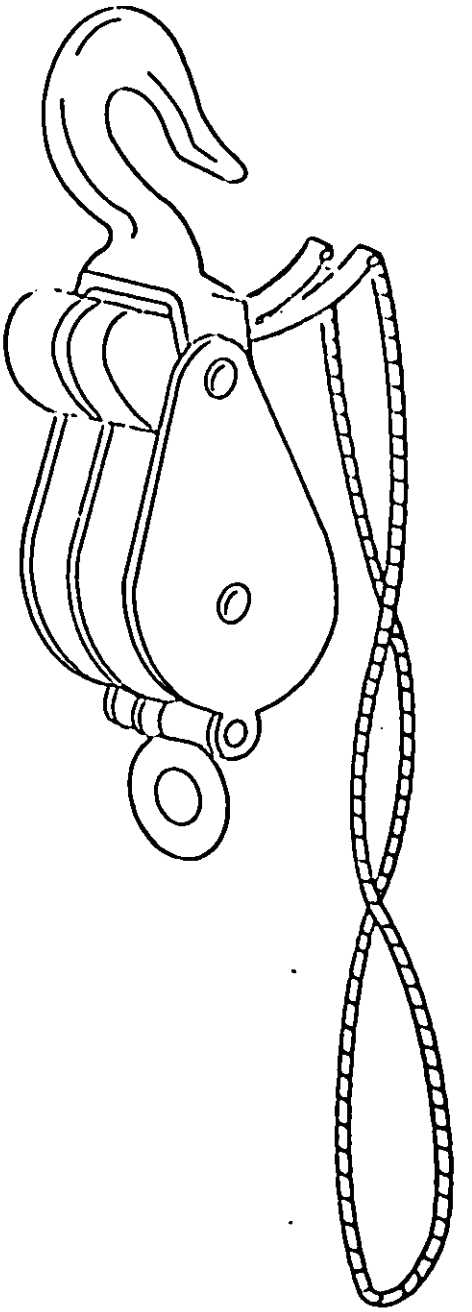


FIGURE 6. Type IV, block with load locking device.

X-4117

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GGG-B-490E	Blocks, Tackle (Manila and Nylon Rope)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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